

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	25 JULY 2026
Team ID	
Project Name	CAB BOOKING
Maximum Marks	4 Marks

Functional Requirements:

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through Form, Registration using Google (Gmail), Email Verification
FR-2	User Authentication	Login using Email & Password, Password Encryption, Forgot Password, JWT Authentication
FR-3	Cab Search	Search Available Cabs, View Cab Details, Filter by Vehicle Type, Fare Estimation
FR-4	Cab Booking	Select Pickup & Drop Location, Choose Cab, Confirm Booking, Generate Booking ID
FR-5	Ride Management	Assign Driver, Accept/Reject Ride, Start Ride, Complete Ride, Cancel Ride
FR-6	Payment Management	Online Payment, Payment Confirmation, Payment History, Generate Receipt
FR-7	Booking History	View Current Bookings, View Previous Rides, Download Booking Details
FR-8	Driver Management	Driver Registration, Driver Login, Manage Availability, Update Ride Status
FR-9	Admin Dashboard	Manage Users, Manage Drivers, Manage Vehicles, Manage Bookings
FR-10	Notifications	Booking Confirmation, Ride Status Updates, Payment Notifications, Email Alerts

Non-functional Requirements:

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The application should provide an easy-to-use, responsive, and user-friendly interface for customers, drivers, and administrators.
NFR-2	Security	User passwords must be encrypted using bcrypt , authentication should use JWT , and protected routes must prevent unauthorized access.
NFR-3	Reliability	The system should provide accurate booking information with minimal failures and recover gracefully from errors.
NFR-4	Performance	API responses should be fast, pages should load quickly, and the system should efficiently handle multiple booking requests.
NFR-5	Availability	The application should be available 24x7 , except during scheduled maintenance, ensuring continuous access to users.
NFR-6	Scalability	The system should support increasing numbers of users, drivers, vehicles, and bookings without significant performance degradation.
NFR-7	Maintainability	The codebase should be modular, well-documented, and easy to maintain or extend with new features.
NFR-8	Compatibility	The application should work correctly on modern browsers (Chrome, Firefox, Edge, Safari) and different screen sizes (desktop, tablet, and mobile).